VIDEO EMOTION RECOGNITION WITH CONCEPT SELECTION

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ABSTRACT

Understanding video content is a challenging problem in many applications, especially for emotion analysis. Diverse and complicated video contents are the major obstacles in video emotion understanding. In this paper, we propose a modality fusion framework to combine concept and content features from action, scene and object models. We conduct concept selection to investigate the relations between high-level concept features and emotions. The discriminative concepts play important roles in emotion recognition. Fusing different modality of content features further improve the performance. The extensive experiments show the state-of-the-art results on two challenging video emotion benchmarks.

Index Terms— Video emotion recognition, deep learning, concept selection

1. INTRODUCTION

The rapid growth of internet and mobile devices creates a great demand for understanding visual media content [1]. Many researchers have explored video understanding tasks, such as video action recognition and detection [2, 3]. While limited efforts have been focused on analyzing emotions in videos, which is also a significant part in video content understanding. Nevertheless, understanding emotions in user-generated videos has many real-world applications. For example, the accurate advertisement recommendation system can benefit from analyzing emotions in the main videos.

Inspired by emotion theories such as Plutchik's wheel of emotions [4] and Ekman's six basic emotions [5], researchers have studied understanding emotions in images extensively in last several years. Although some advances have been made on sentimental analysis on images [6–8], understanding video emotions in user-generated videos still remains challenging. Compared to text and images, video content is more diverse and complicated. Only using low-level visual content features are not enough to bridge the affective gap. And emotions in videos are often triggered by multiple clues, including actions, background scenes and objects rather than still images.

To tackle these challenges, we propose a novel deep learning framework to combine action, scene and object features in emotion recognition using both concept and content features. The classification scores of different categories in different base models are regarded as *concept features*. However, as we observe, some concepts are occurred in multiple emotions, such as some object concepts like tables, cups. The most discriminative high-level concepts help us connect videos to more abstract emotions. Meanwhile we also take fully-connected features as *content features*. The content features from different modalities can provide more details in the whole videos.

In this paper, we first adopt selection strategies based on Bayes' theory to filter out the non-discriminate object, scene and action concepts. Only with a few concepts, we can achieve promising results. Secondly, since the content in videos also preserves significant information, the final video representation also take content features as consideration. Extensive experiments on benchmark datasets Ekman and YouTube are conducted to evaluate our proposed framework.

Our contributions can be summarized as follows.

- We adopt concept selection strategies to identify discriminate concepts in emotion recognition. The concept selection makes our framework more accurate and efficiency.
- We explore different modality of content features in emotion recognition, including action, scene and object features.
- We conduct comprehensive experiments on benchmark video emotion datasets and achieve the state-of-the-art results.

2. RELATED WORKS

2.1. Emotion Recognition on Images

There has been significant progress made on the emotion recognition in images. The authors in [9] performed emotion

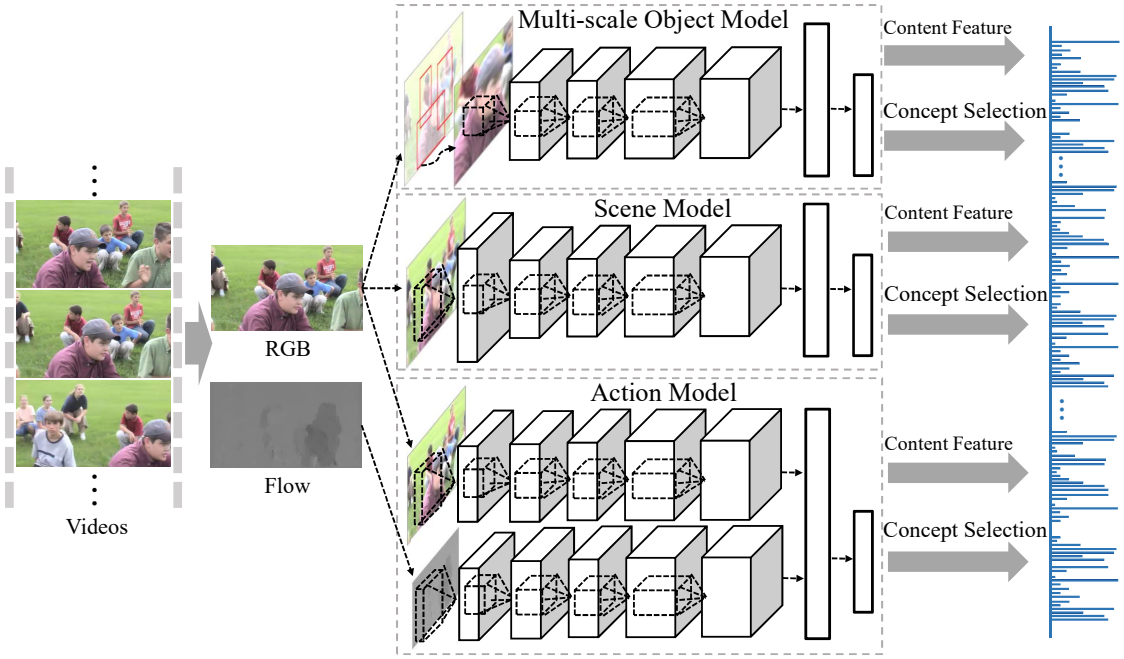

Fig. 1. The whole pipeline of our proposed framework. We first extract static frames and optical flow image from videos. The RGB frames are fed into object model, scene model and action model. While optical flow images are used in action model. Then we perform concept selection to choose high-level features. The final video representations are consists of content features and concept features in different modalities.

recognition using holistic image features. In spite of image features, Machajdik et al. investigated texture, composition, face and skin features inspired by psychology and art theory [10]. Moreover, Lu et al. [6] explored shape in images and how it affected viewers' emotions. The shape features demonstrated highly complementary with low-level visual features.

More recently, convolutional neural network has been widely used in many computer vision tasks as well as emotion recognition. Borth et al. proposed a large scale of visual sentiment image dataset, which contains 1200 adjective-noun pairs. Both an SVM classifier and deep learning network have been used and compared to recognize sentiment concept in images [11, 12]. Furthermore, You et al. [7] proposed a progressively training framework to remove poorly performed images and train CNN using remaining images. While Xu et al. [8] introduced a transfer framework to utilize large-scale CNN which was design to perform object recognition. Those methods using deep neural network showed more effective than hand-crafted visual features.

2.2. Emotion Recognition on Videos

In early studies, most of the works of video emotion recognition have been focused on movies. Kang [13] for the first time proposed a Hidden Markov Model based on low-level visual features. More recently, joint visual-audio features showed promising results on benchmark movie datasets [14, 15].

User-generated videos have more diverse content and low quality, which are more challenging to recognize emotions. Jiang et al. [16] proposed a benchmark video emotion dataset and investigated a large set of low-level and mid-level representations. An image transfer encoding system was proposed by Xu et al. [17] to generate video representations. The deep features have been systematically investigated for emotion recognition. Chen et al. [18] further explored a context fusion framework to integrate high-level semantic features to narrow the affective gap. More recently, Zhang et al. [19] using kernel transform features on emotion recognition.

While the previous methods have not fully considered multi-modalities of emotions as well as the relations between concepts and emotions in videos. In our work, among different concepts of multi-clues, we investigate which concept has more influences on emotions in the entire video. Furthermore, we propose a fusion network, which combines concept and content features in action, scene and object modalities.

3. FRAMEWORK

In this section, we will introduce our proposed framework, which consists of two parts. Figure 1 shows the overview of our method. First, we will introduce the conception selection strategy, and then we will describe the multi-modal feature fusion procedure.

3.1. Concept Selection Strategy

Intuitively, emotion is a high-level concept conveyed by videos, which is closely related to high-level concepts of object, scene and action. The concepts used in our paper are final classification scores generated from different models. We will discuss the detail of models in Section 3.2. We aim to find a way to select most discriminative concepts for distinguishing different emotions.

Inspired by using concepts in scene and event detection [20, 21], we define a criterion that helps us select most discriminative categories to improve emotion recognition. Naturally, since these universal concepts do not relate to emotion classes equally, the first clue indicates that the selected concept should only present in a small number of emotion classes. While there are also some co-occurrence concepts contribute to the prediction of emotion classes, either of them can respond to a specific emotion. As another clue, we want to provide greater diversity of the concept subset, which means the concepts in the chosen subset should have low correlation with others.

Formally, we can define our training video set as

$$Tr = \{(V_i, \boldsymbol{S}(V_i), z_i)\}_{i=1,\cdots,n},$$

where the ith video $V_i = \{\mathbf{v}_{i,1}, \cdots, \mathbf{v}_{i,n_i}\}$ has n_i frames, with the concept scores averaged in each videos $S(V_i)$. z_i is the class label of video V_i from the set of training labels Z and n is the total number of training videos.

In order to get the distribution of different concept classes presenting in training datasets, we then estimate the conditional distribution $p(c|e)$ as follows:

$$p(c|e) = \frac{1}{n_e} \sum_{i:z_i=e} S_i, \tag{1}$$

where $p(c|e)$ is the conditional distributions of concept class c when given emotion class e. n_e represents the number of videos belonging to the eth emotion class(i.e., $n_e = \sum_{i=1}^{N} \mathbb{I}(y_i = e)$). The average score of videos from the same emotion class is utilized to estimate the conditional distribution given a specific emotion category. We also estimate the prior probability of emotion *e* as follows:

$$p(e) = \frac{N_e}{N}. \tag{2}$$

To further investigate the relationship between concepts and the whole emotion video dataset, we define the marginal distributions as:

$$p(c) = \sum_e p(c|e)p(e). \tag{3}$$

Based on the selection rules, we regard this problem as node selection on a fully-connected graph, where each node represents a concept and each edge responses to the correlation between the connected nodes. Meanwhile, each node is associated with a hidden variable $h_i \in \{0, 1\}$, which defines whether the concept is selected or not in the final classification. Thus, we want to minimize the energy function given a set of concept classes $\mathbf{h} = \{h_i\}_{i=1}^{N}$ as follows,

$$E(h) = \sum_{i=1}^{N} \phi(h_i) + \lambda \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} \psi(h_i, h_j), \tag{4}$$

where $\phi(h_i)$ represents the cost of selecting the ith class, $\psi(h_i, h_j)$ is regarded as the cost of having both i^{th} class and j^{th} class. $\sum_{i=1}^{N} h_i = K$ and K is the hyperparameter for the number of selected categories.

In order to ensure the discriminative of selected classes, the score of concepts should only be high for a small subset of emotions. Therefore, entropy is a natural measure to verify this of a probability distribution. The conditional entropy $H(E|c)$ is used to represent the discriminative of the cth concept, and it is formally defined as,

$$H(E|c) = -\sum_e p(e|c) \log_2 p(e|c), \tag{5}$$

where $p(e|c)$ is the conditional emotion distribution given a specific concept class, which can be computed from Eqs.(1) and Eqs.(3) using Bayes' formula:

$$p(e_i|c) = \frac{p(c|e_i)p(e_i)}{\sum_{i=1}^{E} p(c|e_i)p(e_i)}. \tag{6}$$

Thus, $\phi(h_i) = H(E|i)$ given $h_i = 1$.

As for considering correlation among different concepts, the concepts should have similar conditional probabilities $p(e|c)$ if they predict similar emotions. For example, $\psi(h_i, h_j) = \langle p(e|i), p(e|j) \rangle$ if $h_i = 1$ and $h_j = 1$.

For computational convenience, a greedy strategy is used to pick the concept with the lowest conditional entropy. Then we update the correlation of the remaining concepts with selected ones. The concept class with the minimum cost of discriminative and diversity capacity will be chosen next. The iteration is repeated until K concepts are selected. This strategy is used in three models separately.

3.2. Multi-modal Feature Extraction

Emotions are evoked by multiple clues, such as background, actions and other objects in the videos. In this section, we introduce action features, scene features and multi-scale object content features to further improve emotion recognition.

Action Features. Human actions are highly related to emotions. For example, *hugging* and *kissing* often occur with positive emotions, while *crying* may occur with negative emotions.

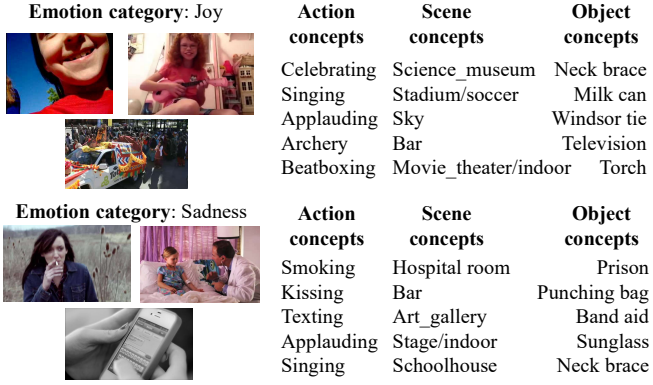

Fig. 2. Visualize emotion categories with corresponding top 5 concept responses(i.e. $p(c|e)$) in VideoEmotion dataset.

More specifically, We take advantage of the large-scale human action recognition dataset Kinetics [22], which covers 400 human action classes in our daily lives. We extract features from the Temporal-Segment-Network [23], which is pre-trained on Kinetics using Inception-V3 [24]. Both RGB and flow features are extracted from the last global pooling layer with 2048 dimension.

Scene Features. Emotion is also associated with the background in videos. Different scenes have different impact on viewers' emotions. Therefore, we use ResNet-50 [25] pre-trained on Place365 [26], which contains 365 scene categories. The 2048 dimensional features before softmax layer are extracted as scene representations.

Object Features. Furthermore, objects in videos often contain significant clues for emotion recognition. However, existing work usually performs image classification on the global level of each frame, which will neglect small size of objects. In our proposed framework, we first use RPN to extract object patches and then feed into ResNet pre-trained on ImageNet. For more detail, we use YOLO-v3 [27] pretrained on COCO dataset [28] as the RPN detector, which includes 80 common objects classes. The 2048 dimensional of fully-connected layer features in ResNet are regarded as object content features.

4. EVALUATIONS

4.1. Dataset & Settings

In this section, we first introduce datasets we use and some experiment settings. Then we perform comprehensive evaluations on two benchmark datasets, including using concept selection and multi-modal features.

Video Emotion Dataset [16]. Inspired by Plutchik's emotion wheel, the Video Emotion dataset contains 1101 videos labeled with 8 basic emotions. All the videos are downloaded from Flickr and YouTube with an average duration of 107 seconds. Following the setting of [16], we perform our experiments on 10 train/test splits and report the mean accuracy of the ten splits.

Table 1. The accuracy of using concept selection features.

Method	VideoEmotion	Ekman
Object	40.05	45.11
Scene	45.62	49.02
Object + Scene	46.38	49.76
Action (RGB)	47.37	49.02
Action (Flow)	41.40	45.11
Action (RGB+Flow)	48.93	53.42
Object + Scene + Action	**50.82**	**54.40**

Table 2. The accuracy of using multi-modal features. OSA indicates the results of Object + Scene + Action in Table 1.

Method	VideoEmotion	Ekman
OSA	50.82	54.40
+Object_fc	50.88	54.65
+Scene_fc	50.96	55.26
+Action_fc	**51.48**	**55.62**

Ekman Dataset [29] is a larger video emotion dataset for understanding user-generated videos. It consists of 1637 videos annotated with Ekman's 6 basic emotions. Each class has a minimum number of 221 videos collected from video sharing website. We report the classification accuracy using 819 training videos and 818 testing videos, which is the same setting in [29].

Experiment Settings. In order to simplify computations, we use the average features of all the frames as video representations. The linear SVM and RBF kernel are used to perform emotion recognition. The kernel fusion is performed when fusing different modalities of features.

4.2. Concept Selection

We first try to visualize the concept responses of different emotion categories. Figure 2 shows some example frames of *Joy* and *Sadness* in VideoEmotion dataset. Some concepts are closely related to emotions, such as celebrating in *Joy* and smoking in *Sadness*. While there also many concepts are corresponding with more than one emotions. Besides, some concepts are related to each other, such as applauding and singing. Either of these related concepts can represent a specific emotion. Thus, as we describe above in Sec. 3.1, we choose the discriminative and diverse concepts occurred in a small number of emotion classes and with low correlation with others.

We evaluate the effect of concept selection strategy on emotion recognition using different modality features. In Figure 4, it clearly shows that using a few concepts (e.g. 200

Table 3. The accuracy of comparing with baseline methods. For fair comparison, the results only consider visual features.

Method	VideoEmotion	Ekman
Hand-crafted SVM	41.90	-
ITE	43.80	50.90
Content Fusion Network	50.60	51.80
Kernelized Feature	49.70	54.40
Concept Selection	50.82	54.40
Multi-modal Fusion	**51.48**	**55.62**

concepts) can also achieve promising results. Compared with randomly choosing some categories, our results get more improvement in almost all circumstances, especially when choosing a small amount of concepts. This provides a more efficient way to reduce the dimension of features to perform emotion recognition.

4.3. Multi-modal Features

In this subsection, we evaluate different modality of features and the fusion of features. Table 1 reports the results of using concept selection features. We empirically choose 250 categories for scene and action concepts, 400 for object concepts. Among the three kinds of clues, it is clear that action features achieve the best accuracy owing to the motion features. The scene features also show benefit for emotion recognition. This demonstrates emotions convey from videos are related to background environment closely. To further improve the results of our methods, we add features from the last fully-connected layer (as shown in Table 2). These features are related less to specific task and carry more general features. The fusion results also indicate that different modalities of features are complementary with each other in emotion recognition task.

4.4. Compare with State-of-the-art Methods

We also compare with some state-of-the-art methods on the two benchmark datasets as follows: **(1) Hand-crafted SVM [16].** This method explored a large set of low-level features and mid-level features. **(2) Image Transfer Encoding [8].** This paper proposed an emotion recognition framework using auxiliary image dataset to help encode videos. **(3) Content Fusion Network [18].** The authors integrated high-level semantics features into a unified context fusion network. **(4) Kernelized Feature [19].** This paper constructed kernel functions to convert CNN features into kernelized features, which shows effectiveness on emotion recognition.

Table 3 shows the accuracy of baseline methods as well as ours. Although the audio clues are not included in our report, it is easy to extend by our framework. The results show our methods achieve the state-of-the-art results on both concept selection ways and add multi-modal fusion. Compare with baseline methods, the concept selection strategies choose more discriminative categories. Besides, multi-modal fusion takes advantage of different clues in emotion recognition.

5. CONCLUSIONS

We proposed a unified framework with concept selection strategies to perform video emotion recognition. The concepts, including actions, scenes, and objects, are extracted using visual information and part of them are filtered with the concept selection strategy. The multi-modal fusion with fully-connected features is further incorporated to improve the final performance. Experiments on two benchmark datasets are conducted and our concept selection framework achieves the state-of-the-art results for video emotion recognition.

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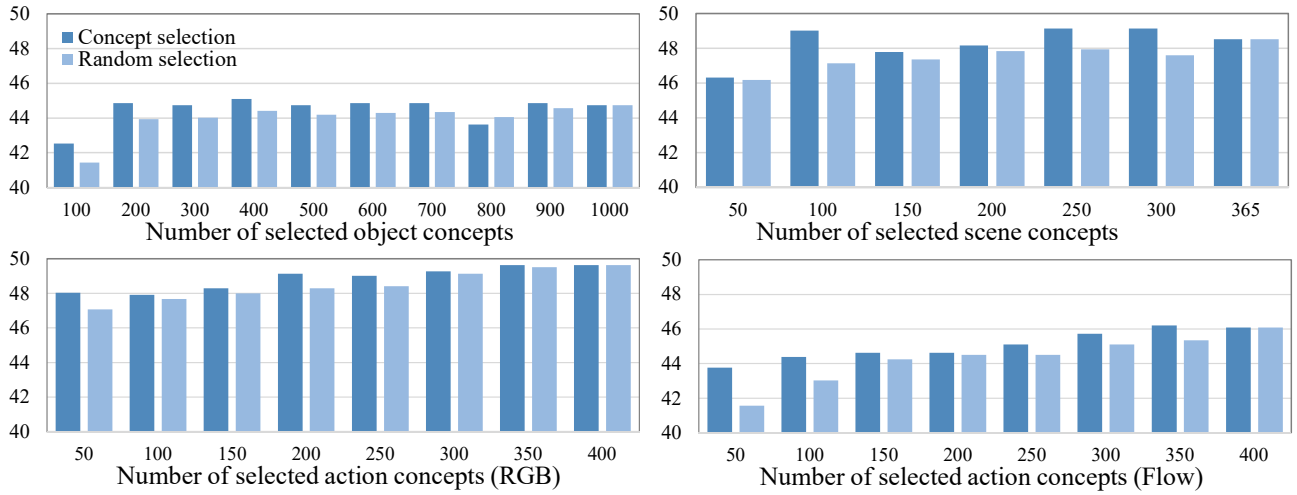

Fig. 3. Results using the proposed concept selection and random sampling on Ekman.

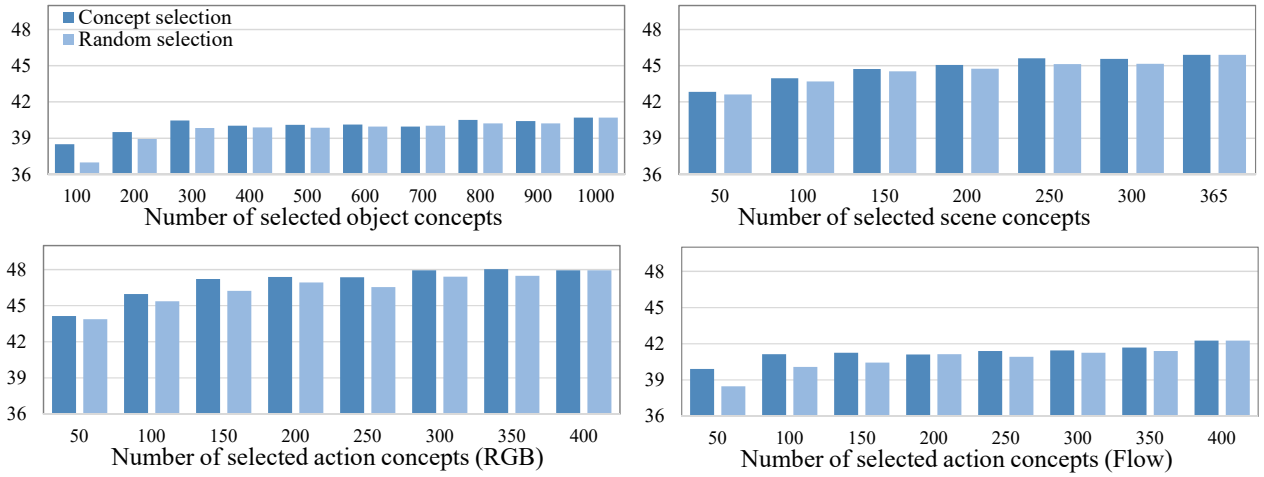

Fig. 4. Results using the proposed concept selection and random sampling on VideoEmotion.

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